VOCATIONAL CLASS 12 — IT | ITES EXAMINATION

SET 2

Time Allowed: 1.5 Hours **Maximum Marks: 30**

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 Marks)

Instructions: Choose the correct option for each question.

1. **Which command is used to add new records to a MySQL table?**
   (a) ADD (b) INSERT
   (c) CREATE (d) UPDATE
2. **What is the correct file extension for Python files?**
   (a) .python (b) .pt
   (c) .py (d) .pyt
3. **Which of the following is a valid variable name in Python?**
   (a) 2variable (b) variablename
   (c) variable_name (d) variable name
4. **In MySQL, which command is used to retrieve data from a table?**
   (a) GET (b) RETRIEVE
   (c) SELECT (d) FETCH
5. **What does RDBMS stand for?**
   (a) Relational Database Management System (b) Random Database Management System
   (c) Reliable Database Management System (d) Remote Database Management System
6. **Which operator is used for exponentiation in Python?**
   (a) ^ (b) **
   (c) // (d) %
7. **In MySQL, which clause is used to sort the result set?**
   (a) SORT BY (b) ARRANGE BY
   (c) ORDER BY (d) GROUP BY
8. **What will be the output of: `print(len("Python"))` ?**
   (a) 5 (b) 6
   (c) 7 (d) Error
9. **Which MySQL function is used to count the number of rows?**
   (a) TOTAL() (b) COUNT()
   (c) NUMBER() (d) SUM()
10. **What is the correct syntax to create a list in Python?**
    (a) list = {1, 2, 3} (b) list = (1, 2, 3)
    (c) list = [1, 2, 3] (d) list = <1, 2, 3>

SECTION B: SHORT ANSWER QUESTIONS (5 × 2 = 10 Marks)

Instructions: Answer any FIVE questions. Each question carries 2 marks.

11. What is the difference between CHAR and VARCHAR data types in MySQL?
12. Explain string slicing in Python with an example.
13. Write a MySQL query to display all records from a table named "Products" where price is greater than 500.
14. What are dictionaries in Python? Write syntax to create a dictionary.
15. Explain the UPDATE command in MySQL with an example.
16. What is the difference between break and continue statements in Python?
17. What is a NULL value in MySQL? How is it different from zero or empty string?

SECTION C: LONG ANSWER QUESTIONS (2 × 5 = 10 Marks)

Instructions: Answer any TWO questions. Each question carries 5 marks.

18. **Explain different types of JOIN operations in MySQL (INNER JOIN, LEFT JOIN, RIGHT JOIN) with examples. Write SQL queries to demonstrate each type.**

19. **What are conditional statements in Python? Explain if, if-else, and elif with examples. Write a Python program to calculate grades based on marks (A: >90, B: 80–90, C: 70–79, D: 60–69, F: <60).**

20. **Explain aggregate functions in MySQL (COUNT, SUM, AVG, MAX, MIN). Write SQL queries demonstrating the use of each function with a "Sales" table containing columns: Product_ID, Product_Name, Quantity, Price.**

21. **What are strings in Python? Explain different string methods (upper(), lower(), strip(), split(), replace()) with examples. Write a program that takes a sentence as input and performs various string operations.**

[End of Question Paper]
Answer Key follows on the subsequent pages.

ANSWER KEY — SET 2

Detailed Solutions & Marking Scheme

SECTION A: MCQ ANSWERS (1 Mark Each)

1. **(b) INSERT** — Adds new records.
2. **(c) .py** — Python file extension.
3. **(c) variable_name** — Valid variable name.
4. **(c) SELECT** — Retrieves data.
5. **(a) Relational DB Management System**
6. **(b) **** — Exponentiation operator.
7. **(c) ORDER BY** — Sorts result set.
8. **(b) 6** — Length of string.
9. **(b) COUNT()** — Counts rows.
10. **(c) list = [1, 2, 3]** — List syntax.

SECTION B: SHORT ANSWERS (2 Marks Each)

11. **CHAR vs VARCHAR:**
    • *CHAR:* Fixed-length data type; pads with spaces to specified length (e.g., CHAR(10)). Faster for static data.
    • *VARCHAR:* Variable-length data type; uses only required space plus length bytes (e.g., VARCHAR(10)). Space-efficient.

12. **String Slicing in Python:**
    Extracting a portion using indices with syntax `string[start:end:step]`.
    Example: `text = "Python"`; `text[0:4]` yields `"Pyth"`.

13. **MySQL Query (Price > 500):**

```
SELECT * FROM Products WHERE Price > 500;
```

14. **Dictionaries in Python:**
    Unordered collection of unique key-value pairs created using curly braces `{}`.
    Syntax: `d = {"key1": value1, "key2": value2}`

15. **UPDATE Command in MySQL:**
    Modifies existing records. Syntax: `UPDATE table SET col = val WHERE condition;`
    Example: `UPDATE Students SET Marks = 95 WHERE Roll_No = 101;` (Always use WHERE to avoid updating all rows).

16. **break vs continue:**
    • `break` : Immediately terminates the loop entirely.
    • `continue` : Skips the current iteration and proceeds with the next loop cycle.

17. **NULL Value in MySQL:**
    Represents missing or unknown data. Different from zero (numeric value) and empty string (string with 0 chars). Checked using `IS NULL` / `IS NOT NULL` .

SECTION C: LONG ANSWERS (5 Marks Each)

18. **JOIN Operations in MySQL:**
    Combine rows from multiple tables based on related columns.
    • **INNER JOIN:** Returns only matching records from both tables.

```
SELECT Students.Name, Classes.Class_Name FROM Students INNER JOIN Classes ON Students.Class_I
```

    • **LEFT JOIN:** Returns all records from the left table and matching from the right.
    • **RIGHT JOIN:** Returns all records from the right table and matching from the left.

19. **Conditional Statements in Python:**
Execute code conditionally using `if`, `elif`, and `else`.
Grade Calculator Example:

```
marks = 87
if marks > 90: grade = 'A'
elif marks >= 80: grade = 'B'
elif marks >= 70: grade = 'C'
else: grade = 'F'
```

20. **Aggregate Functions in MySQL:**
Perform calculations on multiple rows returning a single value: `COUNT()`, `SUM()`, `AVG()`, `MAX()`, `MIN()`.
Example: `SELECT COUNT(*), AVG(Price), MAX(Price) FROM Sales;`

21. **Strings & Methods in Python:**
Strings are immutable character sequences. Key methods:
- `upper()` / `lower()`: Case conversion.
- `strip()`: Removes leading/trailing whitespace.
- `split()`: Splits string into a list.
- `replace()`: Replaces substrings.

Example: `"Python Programming".replace("Programming", "Coding")`

— End of Answer Key —